

Therapeutic Few-Shot: Intelligent Prediction of Treatment Feasibility with Symptom Tokens from Aggregated Data

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Thesis Defense

- Problem & Motivation
- Approach: Symptom Tokens + ETHOS + Few-Shot + Federated
- Experiments & Results
- Key Findings
- Future Work

- **Task:** Predict treatment feasibility as an early ICU outcome proxy.
- **Boundary:** Not autonomous treatment recommendation or replacement of physician judgment.
- **Challenges:**
 - Data scarcity (few labeled examples per disease group)
 - Privacy (data cannot be centralized)
 - Heterogeneous representation (labs, vitals, demographics)
- **Data:** MIMIC-IV v3.1, 2,000 patients, 5 disease nodes

- Mayo Clinic experience: restricted data access (IDs and codes only)
- Question: How to build intelligent prediction when data cannot be centralized?
- **Our approach:** Federated learning + few-shot + compact representations (Symptom Tokens)

MIMIC-IV → Feature Engineering (22→132-D) →

Training: SupCon → KD → Episodic Training →

Fusion: Proto-MAML Encoder + RF → Stacking Meta-Learner → Prediction

- **Multi-phase training:** SupCon pretraining → KD from RF → episodic meta-training
- **Encoder variants:** Grouped-Feature Transformer (V4), GFA-Transformer (V5), FT-Transformer / MLP-Mixer (V6)
- **Stacking:** Calibrated logistic regression fuses few-shot + RF + embeddings

Main Results

Model / Protocol	AUROC	F1	95% CI
Tabular SOTA ceiling	0.748	0.673	–
Stacked BEST (few-shot + RF + ETHOS)	0.746	0.687	[0.699, 0.797]
Few-shot-only BEST	0.670	0.386	[0.612, 0.726]
Strict Exp1 Random Forest	0.762	0.679	–
Strict Exp1 Proposed FewShot	0.611	0.572	–
Federated centralized → federated	0.552 → 0.547	–	gap 0.005

- **Honest strict result:** Proposed FewShot alone is weaker than RF in Exp1.
- **Final deliverable:** stacked fusion reaches AUROC 0.746, close to the tabular ceiling 0.748.
- **Federated result:** privacy-preserving training costs only about 0.005 AUROC in this simulation.

- **RQ1:** Few-shot vs classical? Few-shot alone does not beat RF; stacked fusion closes most of the gap to the tabular ceiling.
- **RQ2:** Optimal K? $K=20$ best in episodic eval (0.554); main model uses $K=5$
- **RQ3:** Federated cost? Small in this simulation: AUROC gap about 0.005.
- **RQ4:** Key components? SupCon pretraining, KD, Proto-MAML, GFA, stacking each contribute; intensity weighting critical in ablation
- **RQ5:** Cross-disease? Partially — underperforms on node_4, node_5

Key Innovations (V0–V6)

- ① **SupCon pretraining**: Creates clustered embeddings for prototypical networks
- ② **Knowledge distillation**: Transfers RF decision boundaries to encoder
- ③ **Proto-MAML**: Test-time adaptation to local patient populations
- ④ **Gated Feature Attention (V5)**: Adaptive per-sample feature selection
- ⑤ **Cross-Attention Prototypes (V5)**: Context-dependent similarity
- ⑥ **Stacking meta-learner**: Fuses few-shot + classical predictions
- ⑦ **Conformal prediction (V6)**: Calibrated uncertainty at 90% coverage

- ① Reproducible treatment-feasibility benchmark on a 2,000-patient MIMIC-IV cohort
- ② Symptom Tokens as clinical representation for episodic learning
- ③ Honest model hierarchy: RF is stronger than raw few-shot; stacked fusion reaches AUROC 0.746
- ④ Federated-ready with small observed cost (AUROC gap about 0.005)
- ⑤ Proto-MAML for hospital-specific adaptation without retraining
- ⑥ GFA and Cross-Attention Prototypes for tabular few-shot learning

- Larger pretraining (full MIMIC-IV, external EHR)
- Combine V5 GFA encoder with V4 stacking
- Cross-institutional federated evaluation
- Conformal prediction for deployment-ready uncertainty
- Prospective clinical validation with clinician feedback

Thank you

Questions?